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**UNIVERSITY OF CAPE TOWN
DEPARTMENT OF ELECTRICAL ENGINEERING**

EEE3094S

Control Systems Engineering

Q1	
Q2	
Q3	
Q4	
Q5	
Q6	
Q7	
Total	

January Examination 2023

TIME: 3 Hours

TOTAL MARKS: 100

INSTRUCTIONS

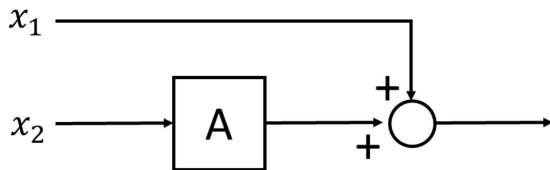
1. Attempt ALL questions.
2. Closed book format.
4. You are allowed one **hand written**, double sided, A4 formula sheet.
3. Hand in all drawings on which you have worked and your formula sheet.
4. Transform tables and Engineering charts are provided at the end of this paper.

Internal Examiners: Prof E Boje, Dr S Shield

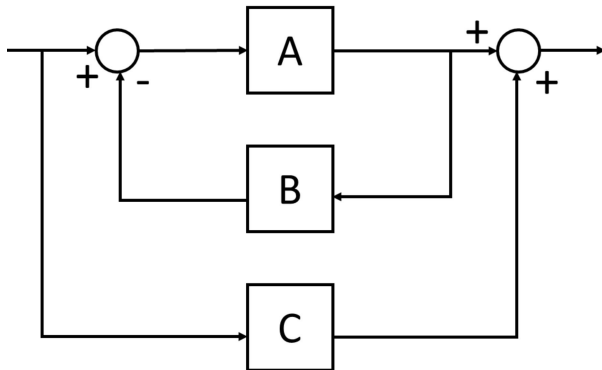
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Question 1 [14 marks]

- a) Draw an equivalent of the following block diagram with the summing junction moved to before block A. Indicate the values of the signals at all nodes (stretches where the value is constant) in terms of x_1 and x_2 to confirm that the systems give the same output. (3)



- b) Simplify the following block diagram into a single block. (3)



- c) State the type number of the system $L(s) = \frac{3}{s(s+2)}$ and give its error constant with respect to a unit ramp input (in a negative gain feedback configuration). (2)

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Question 2 [15 marks]

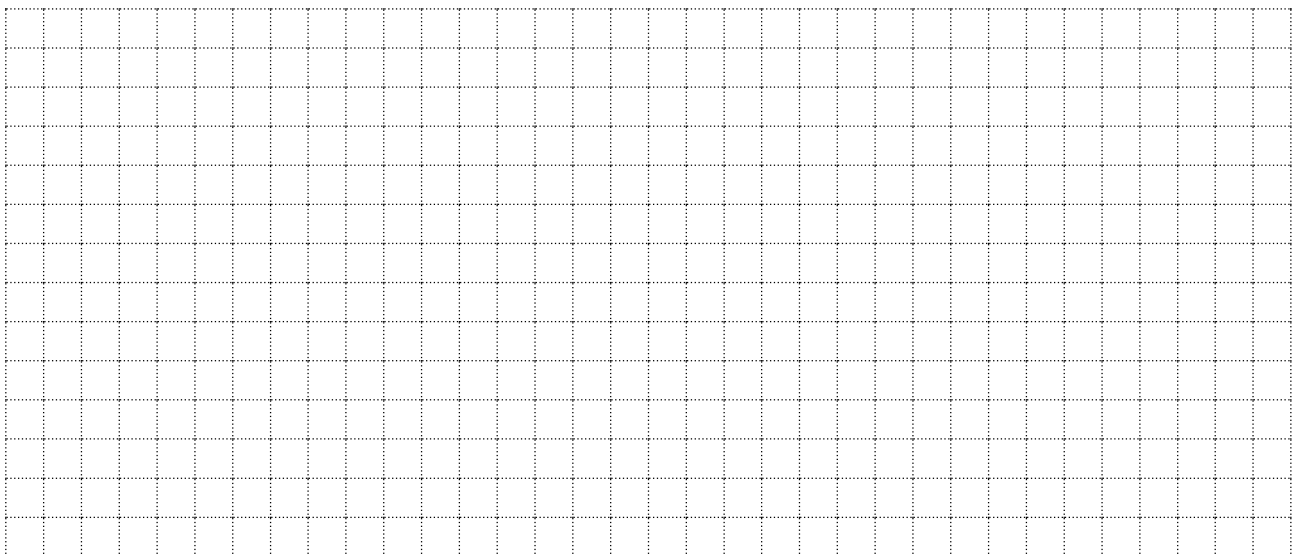
Consider the Bode plot of a system shown in Figure 2.

a) Use the plot to determine the following characteristics of the system:

- i. The gain, A (1)
- ii. The natural frequency, ω_n (2)
- iii. The damping ratio, ζ (2)

b) Determine the damped frequency ω_d , rise time t_r , time to peak t_p , peak magnitude M_p and settling time to within 5% of the final value $t_{5\%}$. (5)

c) Hence, sketch the unit step response of the system, showing the characteristics in (b). (5)



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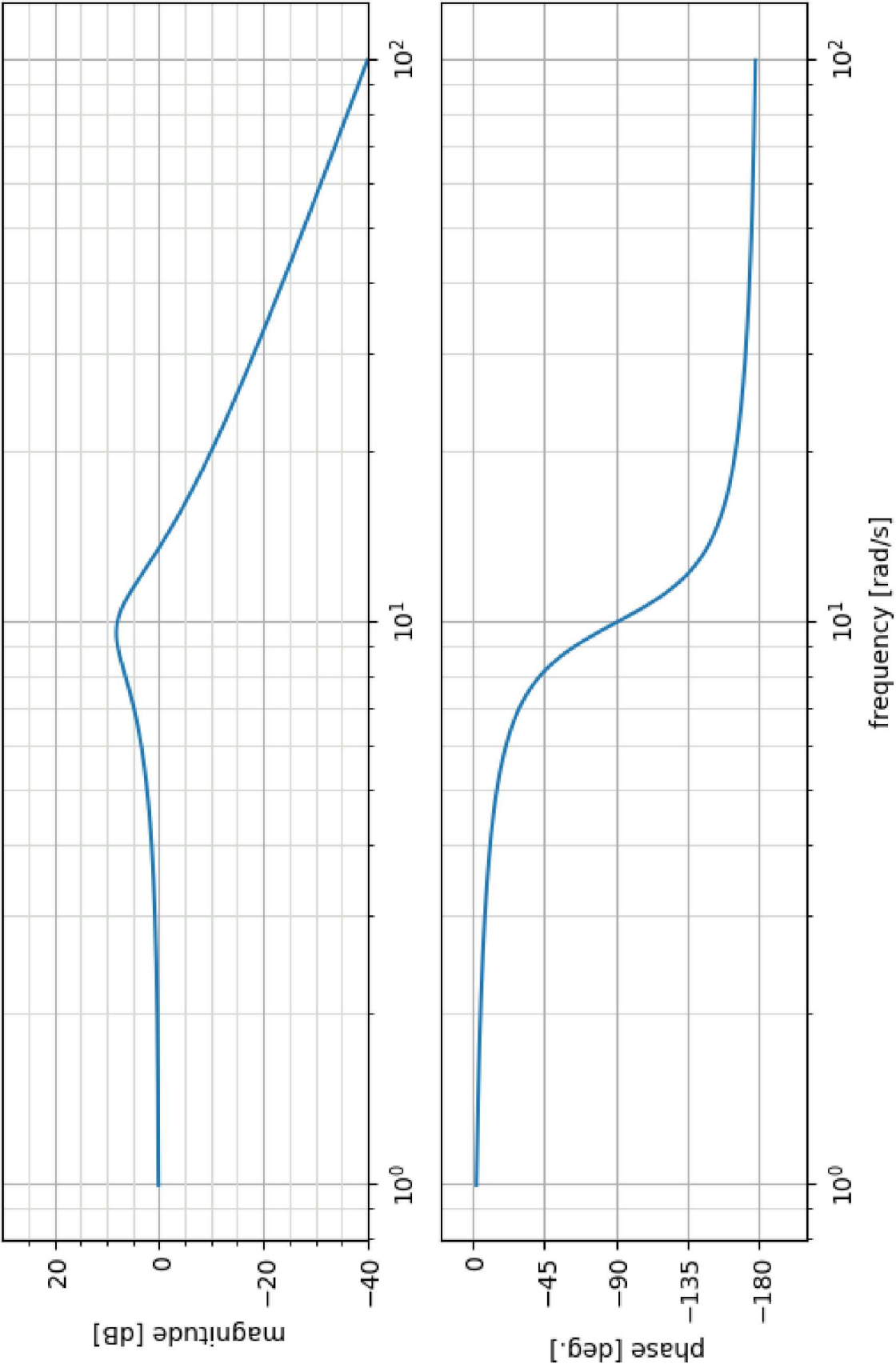


Figure 2: Bode plot

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Question 3 [17 marks]

Consider a system with the transfer function $P(s) = \frac{60}{(s+10)(s+1)(s+3)}$.

- a) Explain why $P_a(s) = \frac{6}{(s+1)(s+3)}$ is a suitable reduced-order model of the system. (You may use this model for the subsequent parts of this question.) (2)

- b) The system is required to track a step reference with 95% accuracy. Your colleague suggests implementing this by putting the system in closed loop with a controller of the form $G_1(s) = \frac{k}{s}$. Explain one advantage, and one disadvantage of this option. You may want to support your answer with a rough sketch of the root locus. (3)

- c) Another colleague suggests a lead compensator, $G_2(s) = \frac{6.51(s+4.5)}{(s+10.5)}$.

- i) Show that this compensator places the dominant poles of the closed-loop system at $-4 \pm j4$. What will the settling time to 5% of the final value, and damping ratio of the system be if this compensator is used? (5)

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QUESTION 4 [18 marks]

Investigate the closed loop stability (with unity gain, negative feedback) of $L(s) = \frac{8(s+6)}{s^2-4s}$.

- a) How many right-hand plane open loop poles does the system have? (1)

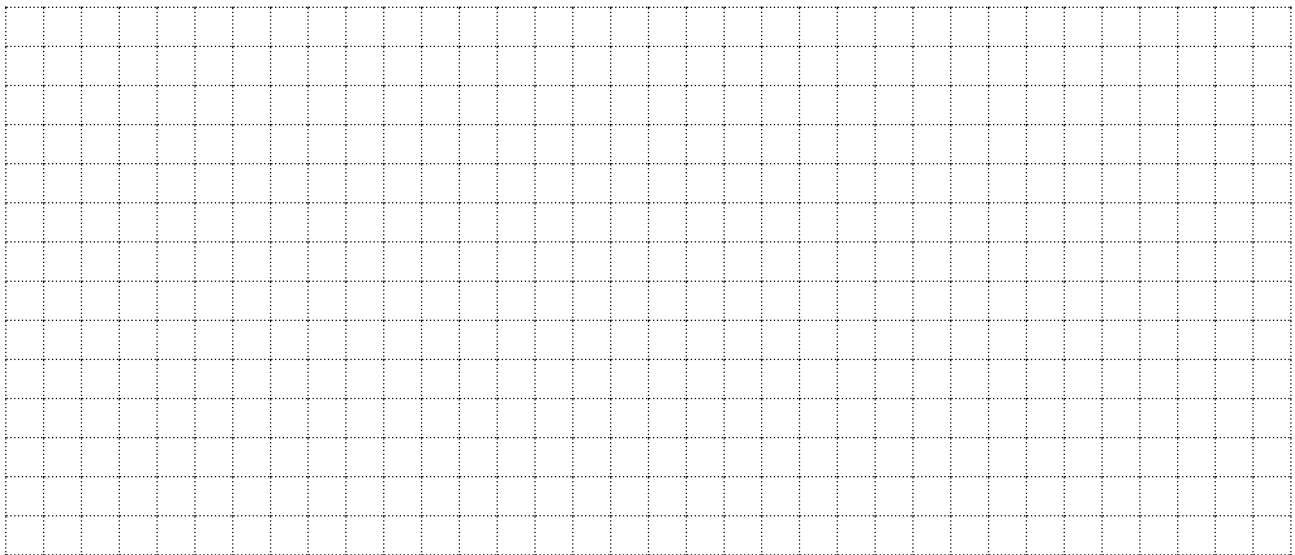
- b) Draw the Bode plot on attached semilog paper, using asymptotic values supported by calculations at $\omega = (5, 7, 9)$ rad/s. Make sure that you get the phase in the correct quadrant. (5)

- c) Hence draw the full Nyquist plot with appropriate scaling. Explain clearly how you deal with singularities on the $j\omega$ axis. (4)

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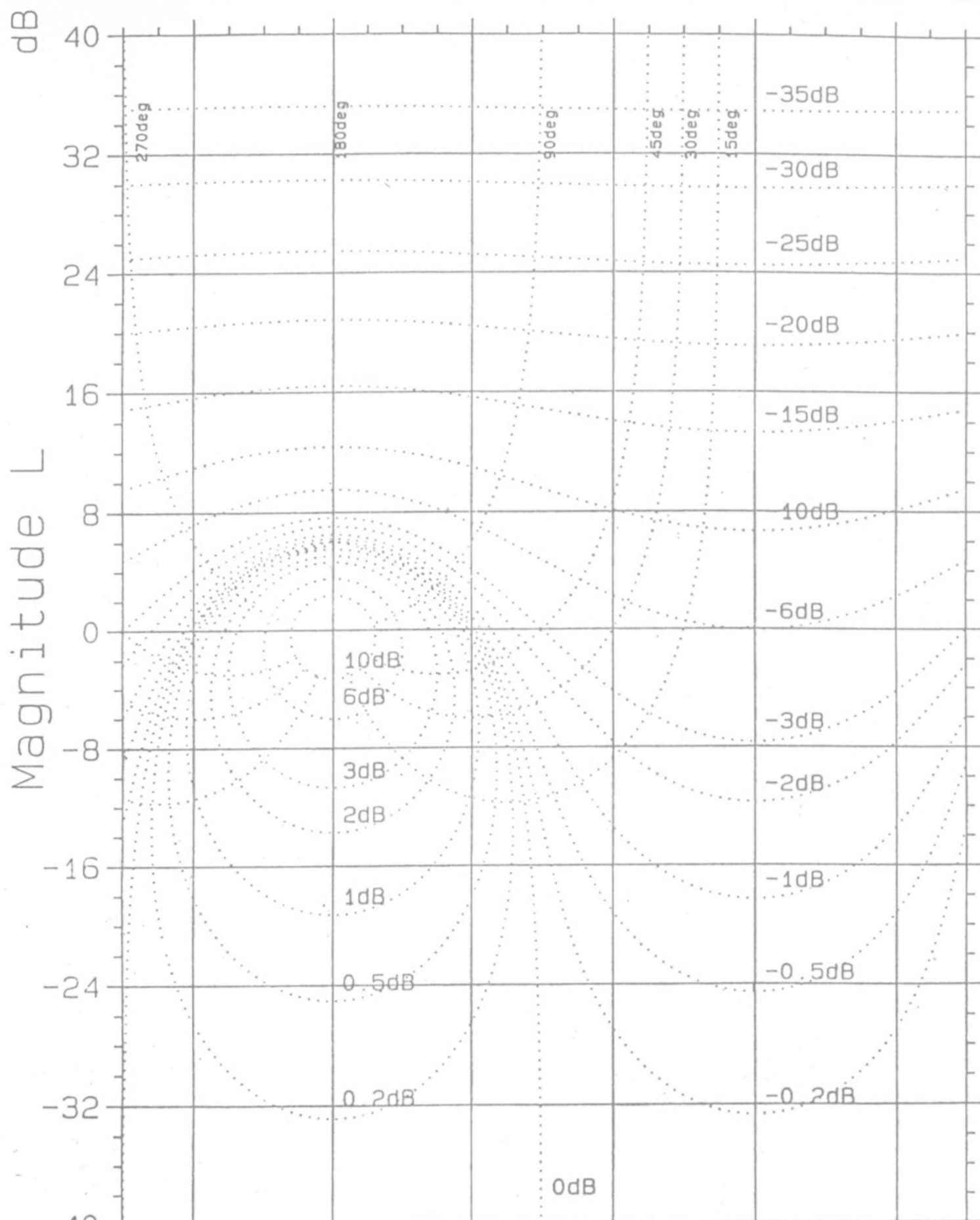
- d) Is the closed loop system stable or not? If it is stable, what are the gain, phase and robust stability margins? If it is unstable, what could be done to stabilise it? (4)

- e) Make a sketch and show that a robust stability margin of γ (arithmetic) implies a phase margin of at least $\phi_m \geq 2 \arcsin \frac{1}{2\gamma}$. Hence find the minimum phase margin corresponding to a specification of $\left| \frac{1}{1+L} \right| \leq 6 \text{ dB}$. (4)



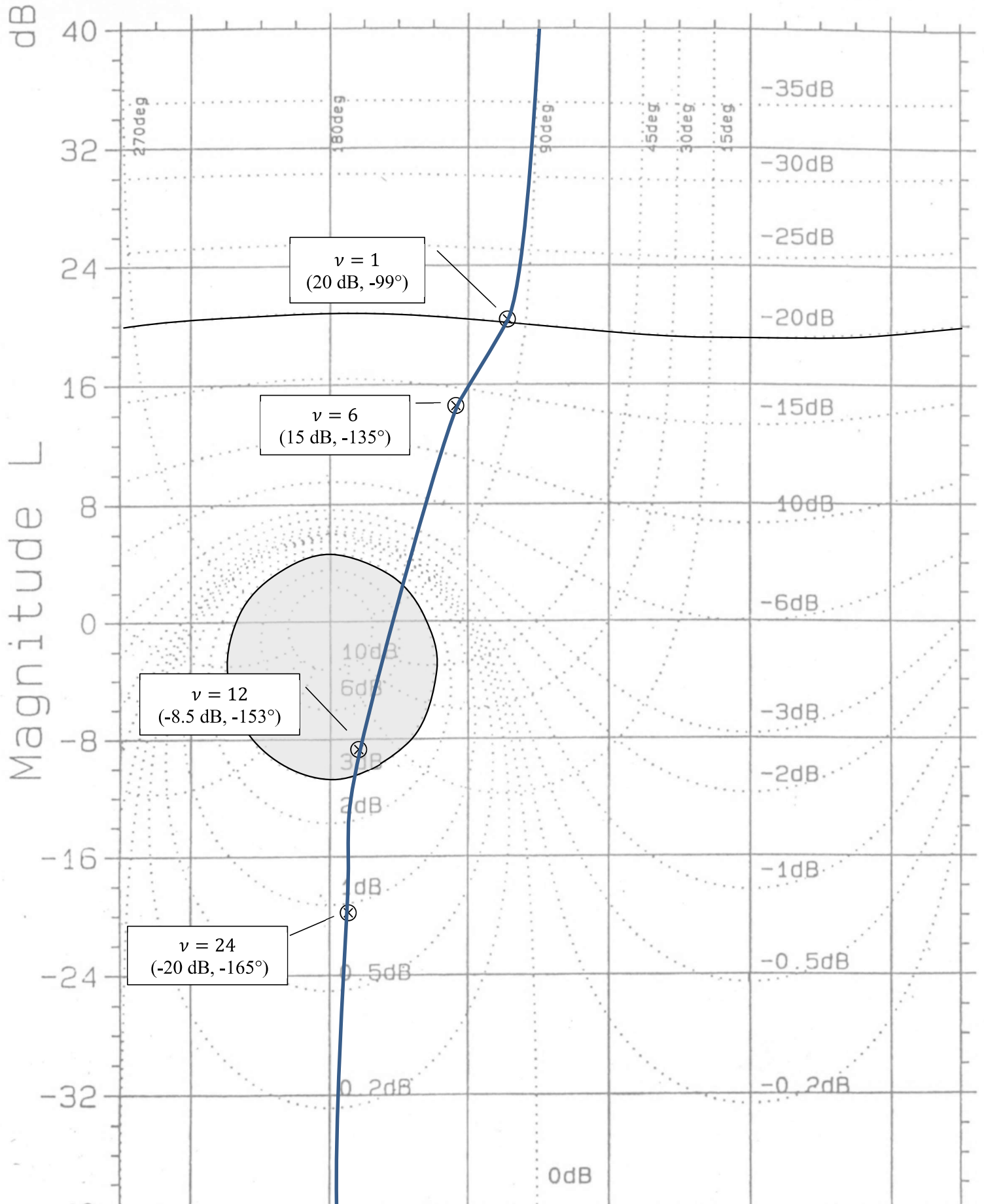
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Inverse Nichols Chart



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Inverse Nichols Chart



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Laplace and Z-transforms.

$Z\{y(iT)\}$	$y(t)$	$L\{y(t)\}$ or $P_s(s)$	$P_z(z) = \frac{z-1}{z} \mathcal{Z} \left\{ \left(\mathcal{L}^{-1} \left\{ \frac{P_s(s)}{s} \right\} \right)_{y=iT} \right\}$
$\frac{z}{z-1}$	$\sigma(t)$	$\frac{1}{s}$	$\frac{T}{z-1}$
$\frac{Tz}{(z-1)^2}$	t	$\frac{1}{s^2}$	$\frac{T^2 / 2(z+1)}{(z-1)^2}$
$\frac{z}{z-e^{-aT}}$	e^{-at}	$\frac{1}{s+a}$	$\frac{(1-e^{-aT})/a}{z-e^{-aT}}$
$\frac{Tze^{-aT}}{(z-e^{-aT})^2}$	te^{-at}	$\frac{1}{(s+a)^2}$	$\frac{(1-e^{-aT}(1+aT))z + e^{-aT}(e^{-at}-1+aT)}{(z-e^{-aT})^2 a^2}$
$\frac{ze^{-at} \sin bT}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$	$e^{-at} \sin bt$	$\frac{b}{(s+a)^2 + b^2}$	$\frac{(n_1 z + n_0)b / (a^2 + b^2)}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$ $n_1 = 1 - e^{-aT} \left(\cos bT + \frac{a}{b} \sin bT \right)$ $n_0 = e^{-aT} \left(e^{-aT} - \cos bT + \frac{a}{b} \sin bT \right)$
$\frac{z^2 - ze^{-at} \cos bT}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$	$e^{-at} \cos bt$	$\frac{s+a}{(s+a)^2 + b^2}$	$\frac{(n_1 z + n_0)a / (a^2 + b^2)}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$ $n_1 = 1 - e^{-aT} \left(\cos bT - \frac{b}{a} \sin bT \right)$ $n_0 = e^{-aT} \left(e^{-aT} - \cos bT - \frac{b}{a} \sin bT \right)$

Property	Time domain	z Transform
Linearity	$au(t) + bv(t)$	$au(z) + bv(z)$
Real Translation	$u(t - kT)$	$z^{-k}u(z)$
Initial Value Theorem	$\lim_{t \rightarrow 0} u(t)$	$\lim_{z \rightarrow \infty} u(z)$
Final Value Theorem	$\lim_{t \rightarrow \infty} u(t)$	$\lim_{z \rightarrow 1} (1 - z^{-1})u(z)$